Haley Chow

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PROFESSIONAL SUMMARY

First-year Biomedical Engineering student with independent research experience in pharmacogenomics, bacterial growth, and statistics. Skilled in experimental design, bioinformatics analysis, and building data tools in Python and JavaScript. Focused on genetics and rare diseases, with a drive to translate research findings into resources that support patients and caregivers.

EDUCATION

Boston University Boston, MA
B.S. in Biomedical Engineering. *Expected May 2030*
Relevant Coursework: Fundamentals of Programming, General Chemistry, Introduction to Engineering

Rowland High School Rowland Heights, CA
Diploma / IB Diploma. GPA: 4.2. *June 2026*
Relevant Coursework: Mathematics: Applications and Interpretation HL, Biology HL, History of the Americas HL, English HL, Mandarin SL, Psychology SL, Theory of Knowledge

PROJECTS

Pharmacogenomic Analysis of CYP2C9 Variants and NSAID Response August 2026 – Present
Same Pill, Different Person: A Pharmacogenomics Study of Celecoxib
- Investigated how CYP2C9 variants (*2, *3) alter celecoxib metabolism using PharmGKB, gnomAD, and CPIC
- Compared variant frequencies across five ancestry groups and translated findings into evidence-based dosing recommendations

IB Mathematics AI HL Internal Assessment: From Peas to Probability January 2026 – March 2026
A Statistical Test of Mendelian Ratios
- Applied three statistical methods to test Mendel's 3:1 and 9:3:3:1 ratios on 15,900+ pea plant offspring; found no significant deviation (all $p > 0.6$)

IB Biology Extended Essay: Antibacterial Effects of Honey and Garlic on E. coli After 48 Hours November 2025 – March 2026
- Designed and executed an original investigation into the antibacterial effects of honey and garlic on E. coli individually and in combination over 48 hours

IB Biology Internal Assessment: Effect of Methionine on E. coli Growth and Protein Synthesis November 2024 – March 2026
- Independently designed and ran an experiment examining whether methionine alters E. coli growth and protein synthesis, analyzing dose-response trends across four trials

GeneScope: DNA Sequence Analyzer July 2026 – Present
Python, Flask, HTML, CSS, JavaScript
- Built a full-stack web app that analyzes DNA sequences and visualizes nucleotide composition
- Implemented sequence algorithms (reverse complement, RNA transcription, protein translation) behind a Python/Flask backend and interactive JavaScript frontend

FinIQ: Zebrafish Research Data Analyzer January 2026 – Present
Python, HTML, CSS, JavaScript
- Developed a research dashboard that imports zebrafish CSV data and compares experimental groups across developmental stages
- Automated statistics, visualizations, searchable tables, and downloadable reports

Axiom: Concept Gap Mapper March 2026 – Present
Python, Whisper, sentence-transformers, NetworkX, D3.js
- Designed and built an AI-powered study tool that maps lecture notes into a concept graph and uses speech and semantic similarity to identify gaps in a student's understanding

The Veniverse (Beyond the Blueprint) November 2024 – Present
UI/UX Design, Python, HTML, CSS, JavaScript
- Built an educational platform to raise awareness of genetic diseases and vascular malformations
- Integrated medical research, 3D anatomical models, and treatment resources for patients and caregivers
- Designed accessible resources connecting patients with specialists and community support

EXPERIENCE

Tennis Coach November 2025 – August 2026
Individual Lessons and JV Boys' and Girls' Teams *Rowland Heights, CA*
- Coached young players in one-on-one tennis lessons
- Coached the JV boys' and girls' tennis teams

Co-Founder December 2024 – October 2025

One Thread at a Time (Heart to Heart) *Rowland Heights, CA*

- Handcrafted comfort pillows, superhero capes, and weighted stuffed animals for pediatric cancer patients while raising awareness of endangered animals
- Partnered with elementary schools to create workshops on animal endangerment and teach lifelong skills

Tutor August 2023 – June 2025

Rowland High School *Rowland Heights, CA*

- Tutored students in Algebra, Calculus, Biology, Chemistry, and History

ACTIVITIES & LEADERSHIP

FCCLA (Family, Career and Community Leaders of America) April 2025 – May 2026

Regional Vice President of Membership

- Spearheaded member recruitment across 18 regional chapters; managed membership data and coordinated growth initiatives organization-wide.

CIF-SS Division IV High School Tennis July 2022 – June 2026

Varsity Captain and CIF-SS Champions for Character Recipient

- Served as captain and three-year starter on the Rowland High School varsity team; organized team workouts and earned the CIF-SS Champions for Character award.

Interact Club, Rotary Club of Walnut Valley August 2024 – June 2026

President

- Led service projects designed to address community needs and encourage student involvement.

Science National Honor Society, Rowland High School October 2025 – June 2026

Secretary

- Organized community service activities and maintained documentation of meetings and attendance.

SKILLS

Research and Lab: Experimental Design, Bioinformatics, Pharmacogenomics, Bacterial Culturing, Aseptic Technique, Serial Dilutions, Microscopy, Colony Counting, Solution Preparation
Data and Analysis: Statistical Analysis, Data Visualization, Chi-Square Testing, Dose-Response Analysis, Variant Frequency Analysis, Descriptive Statistics, Excel
Programming: Python, JavaScript, Flask, HTML, CSS, Java, D3.js, NetworkX, GitHub
Design and Engineering: UI/UX Design, Website Development, Prototyping, Prosthetic Design, Arduino
Languages: English (Native), Mandarin (Conversational)

AWARDS & HONORS

- CIF-SS Champions for Character Recipient
- 6x Principal's Honor Roll
- Hall of Fame Athlete
- 4x Scholar-Athlete Award
- Valle Vista League Top 5